The Coherence-Delay Trade-off

Temporal Geometry of Useful Memory Under Drift

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Abstract

Finite-memory tracking under drift is governed by a coherence-delay trade-off: retaining more past information reduces statistical noise, but it also increases the temporal misalignment between retained evidence and the current distribution. As the target law moves, each additional sample becomes both useful and increasingly stale, so the problem is not simply how much to remember, but how long remembered evidence should remain operationally current.

We study this trade-off under Wasserstein nonstationarity. For the finite-memory averaging class analyzed here, the tracking error splits into a statistical term that decreases with memory and a staleness term that grows with drift. Balancing those two forces yields a **cube-root law**: the **optimal memory horizon** scales as $n^* \asymp (C_K/\zeta)^{2/3}$, while the corresponding **minimum tracking error** scales as $\mathcal{E}_{\min} \asymp C_K^{2/3}\zeta^{1/3}$. In the critical-window regime, we also prove a minimax lower bound, showing that this finite-memory floor is structural rather than estimator-specific.

The experiments separate three empirical layers: horizon geometry, horizon tracking, and predictive payoff. Gaussian drift experiments recover the predicted 'U'-curve and the cube-root scaling. Continuous-drift benchmarks expose a temporal-validity gap: memory can become operationally stale before changepoint evidence is available. Alternating-timescales benchmarks show that the useful horizon moves locally and that contraction is easier than re-expansion. A final gap-cost analysis shows that horizon misalignment induces nonlinear predictive cost, while the transfer from misalignment to error remains method- and regime-dependent.

The resulting picture is theorem-first and operational: useful memory under drift has geometry, temporal coherence is not the same object as changepoint evidence, and the predictive value of horizon control depends on the interaction between drift geometry and memory policy. Viewed through an Age-of-Information lens, the law quantifies how much temporally coherent evidence a finite-memory learner can preserve while tracking a changing world.

Keywords: tracking under drift; useful memory; temporal coherence; Wasserstein distance; Sinkhorn divergence; concept drift; Age of Information; adaptive memory.
MSC 2020: 93D05, 93D30, 68P99.

1 Introduction

Adaptive systems rarely track a fixed target. In streaming estimation, monitoring, and online decision-making, the data-generating distribution itself drifts over time. A learner with finite memory must therefore solve a basic but easy-to-miss problem: how much of the past should still count once the world that produced it has started to move away?

That question creates a structural trade-off. Using more past data reduces sampling noise, but it also increases the age of the information being used to track the present. Under drift, memory is simultaneously helpful and harmful: short windows are fresh but noisy, while long windows are stable but stale. The central claim of this paper is that this tension defines a coherence-delay trade-off and induces a quantitative law for finite-memory tracking under Wasserstein nonstationarity.

For the finite-memory averaging class analyzed here, the tracking error splits into a statistical term that decreases with memory length and a staleness term that increases with drift and effective data age. Optimizing that balance yields a cube-root law: $\mathcal{E}_{\min} \asymp C_K^{2/3}\zeta^{1/3}$ with $n^* \asymp (C_K/\zeta)^{2/3}$. The law is exact for the averaging class studied in the paper. It is not asserted as a theorem for every adaptive estimator. What the paper does show beyond that class is a minimax lower bound for window-restricted estimators in the critical-window regime, establishing that the finite-memory floor is structural rather than a quirk of one specific rule.

This viewpoint also gives the paper its interpretive layer. The limiting factor is the age of the information that finite memory forces the estimator to retain, alongside the usual finite-sample cost. The result can therefore be read as a temporal-coherence law: the learner pays for variance reduction by carrying older evidence, and optimal tracking occurs at a memory horizon that balances statistical stability against operational staleness. In Age-of- Information terms, timeliness belongs to the estimation problem itself.

The empirical story is more nuanced than a dominance claim. Some regimes reward a short fixed window as a strong operational compromise, even when the local oracle horizon moves over time; what the cube-root regulator adds is a way to track horizon validity, not a guarantee of lower error everywhere. The paper therefore separates horizon geometry, horizon tracking, and predictive payoff.

The empirical section follows the same logic in four steps. Continuous-drift benchmarks recover the 'U'-curve and expose the gap between temporal validity and detection. The temporal-validity benchmark measures the delay between cap activation and ADWIN detection, the alternating-timescales benchmark exposes horizon instability, and the gap-cost benchmark quantifies how misalignment translates into predictive cost. Sinkhorn-based tracking scores serve as the measurement layer, and the cube-root cap shows how the same law can act as an online regulator as well as a retrospective explanation.

The scope of the paper is deliberately narrow. We study finite-memory tracking under drift, a lower bound for restricted estimators, and the associated temporal-coherence interpretation, together with experiments chosen to test that story. The paper asks a single question: what tracking accuracy is possible when the learner's memory is finite and the target distribution keeps moving?

2 Useful Memory Geometry

Under continuous drift, memory is not only a statistical resource but also a temporal liability. The core question is therefore not whether one can react to change, but how long a sample should remain operationally useful before it becomes stale.

Definition 2.1 (Lag-Inclusive Estimation Error)**.** For a finite-memory estimator with effective window size n and drift rate ζ, we write the expected tracking error as

$$\mathcal{E}(n) \;=\; C_K\, n^{-1/2} \;+ \tfrac{1}{2}\,\zeta\, n,$$

where the first term is the finite-sample statistical cost and the second term is the staleness cost induced by retaining old information.

The two terms in definition 2.1 pull in opposite directions. Increasing n reduces variance but raises the age of the retained evidence. That tension is the basic coherence-delay trade-off that governs useful memory under drift.

Proposition 2.2 (Optimal Window Under Drift)**.** *Under definition 2.1, the error $\mathcal{E}(n)$ is minimised at*

$$n^* \;=\; \left(\frac{C_K}{\zeta}\right)^{2/3},$$

with minimum value

$$\mathcal{E}_{\min} \;=\; \tfrac{3}{2}\, C_K^{2/3}\, \zeta^{1/3}.$$

Proof. Differentiate $\mathcal{E}(n)$ with respect to n and solve $d\mathcal{E}/dn = 0$. The unique positive critical point gives the stated window. Substituting back yields the minimum value. ☐

Corollary 2.3 (EWMA Analogue)**.** *For exponential weights $w_j = \alpha(1-\alpha)^j$, the effective lag is $\tau_\alpha = (1-\alpha)/\alpha$ and the effective sample size scales as $n_{\text{eff}} \asymp 1/\alpha$. Balancing the same statistical and staleness terms therefore yields the same cube-root exponent in the effective memory scale.*

Definition 2.4 (Effective Memory Age)**.** Let an estimator place weights $w_0, \ldots, w_{n-1}$ on the most recent observations, with $\sum_j w_j = 1$. Its *effective memory age* is

$$A(w) \;=\; \sum_{j=0}^{n-1} j\, w_j,$$

and its corresponding freshness proxy is the monotone transform $F(w) = 1/(1 + A(w))$. The role of $F(w)$ is purely geometric: it records how old the retained evidence is before any local drift scale is specified. For a uniform window, $A(w) = (n-1)/2$.

Definition 2.5 (Temporal Coherence)**.** For an estimator with weights w at time t, define its *operational staleness*

$$S_t(w) \;=\; \sum_{j=0}^{n-1} w_j\, W_2(\mathcal{P}^*(S, t-j), \mathcal{P}^*(S, t))\,.$$

Its *temporal coherence* is the inverse alignment score

$$C_t(w) \;=\; \frac{1}{1+S_t(w)}.$$

If the target path is locally W_2-Lipschitz at rate ζ_t, in the sense that $W_2(\mathcal{P}^*(S,t-j),\mathcal{P}^*(S,t)) \le \zeta_t j$ for the relevant lags, then

$$S_t(w) \;\le\; \zeta_t A(w), \qquad C_t(w) \;\ge\; \frac{1}{1+\zeta_t A(w)}.$$

In that sense, $F(w)$ is the rate-free freshness proxy, while $C_t(w)$ measures operational alignment with the current distribution.

The quantity $A(w)$ gives a compact description of the temporal geometry of a finite-memory estimator. Short windows preserve temporal coherence at the cost of variance; long windows reduce variance at the cost of staleness. The cube-root horizon marks the point where that exchange stops improving. For a uniform window under approximately constant local drift, the induced staleness scales as $S_t(w) \approx \zeta(n-1)/2$, which is the same age term that appears in definition 2.1 up to constants.

Proposition 2.6 (Finite-Memory Floor)**.** *Let the target trajectory satisfy a W_2-Lipschitz drift bound with rate ζ, and restrict attention to estimators that only use the last m samples. Then, in the critical-window regime, the minimax tracking error over that class cannot beat the same variance-plus-staleness scaling:*

$$\inf_{\hat{\mathcal{P}}_m} \sup_{\mathcal{P}^*} \mathbb{E}\, d\big(\hat{\mathcal{P}}_m, \mathcal{P}^*\big) \;\gtrsim\; C_K\, m^{-1/2} + \tfrac{1}{2}\,\zeta\, m.$$

Consequently, the cube-root horizon is not merely the optimum of one estimator; it is the structural floor of finite memory on this drift class.

Proof. Write $R_m^* := \inf_{\delta_m} R_m(\delta_m)$. We exhibit two independent obstructions. The first is the ordinary finite-sample estimation floor, obtained by restricting to the constant-path subclass $\mu_s \equiv \theta$, which is contained in the admissible drift class. For the two hypotheses $\theta_\pm = \pm\sigma/(2\sqrt{m})$, the induced product-Gaussian laws have Kullback–Leibler divergence

$$\mathrm{KL}(P_+,P_-) = \frac{1}{2\sigma^2}\sum_{j=1}^{m}(\theta_+-\theta_-)^2 = \frac{1}{2\sigma^2} m \left(\frac{\sigma}{\sqrt{m}}\right)^2 = \frac{1}{2}.$$

For any estimator δ_m, the plug-in test $\psi = \mathbf{1}\{\delta_m \ge 0\}$ must err whenever the estimate falls on the wrong side of the origin; on that event the endpoint error is at least $\sigma/(2\sqrt{m})$. By Le Cam's two-point lemma and Pinsker's inequality, the average testing error is at least

$$\frac{1-\mathrm{TV}(P_+,P_-)}{2} \ge \frac{1-\sqrt{\mathrm{KL}(P_+,P_-)/2}}{2} \ge \frac{1}{4}.$$

Hence

$$R_m^* \;\ge\; \frac{\sigma}{8\sqrt{m}},$$

so one valid explicit choice is $c_0 = 1/8$.

The second obstruction is local drift. Fix any $1 \le h \le m$ and define

$$\beta_h := \min\Big\{\zeta,\ \frac{\sigma}{4h^{3/2}}\Big\}, \qquad \mu^{\pm,h}_{t-j} := \pm\,\beta_h\,(h-j)_+, \qquad 0 \le j \le m-1,$$

where $(u)_+ := \max\{u, 0\}$. Each path belongs to the admissible class because its slope magnitude is at most $\beta_h \le \zeta$. The endpoint gap is

$$|\mu^{+,h}_t - \mu^{-,h}_t| = 2\beta_h h.$$

The corresponding Gaussian product laws satisfy

$$\mathrm{KL}(P_{+,h}, P_{-,h}) = \frac{1}{2\sigma^2}\sum_{j=0}^{m-1}\big(2\beta_h(h-j)_+\big)^2 \le \frac{2\beta_h^2}{\sigma^2}\sum_{r=1}^{h} r^2 \le \frac{2\beta_h^2}{\sigma^2}h^3 \le \frac{1}{8}.$$

Applying the same reduction from estimation to testing, now with endpoint error threshold $\beta_h h$, and using Le Cam together with Pinsker gives average testing error at least

$$\frac{1-\mathrm{TV}(P_{+,h}, P_{-,h})}{2} \ge \frac{1-\sqrt{\mathrm{KL}(P_{+,h}, P_{-,h})/2}}{2} \ge \frac{3}{8}.$$

Therefore

$$R^*_m \ \ge\ \frac{3}{8}\,\beta_h h \ \ge\ \frac{3}{32}\,\min\big\{\zeta h,\ \sigma h^{-1/2}\big\}$$

which is again a fully explicit numerical lower bound. Since h was arbitrary,

$$R^*_m \ \ge\ \frac{3}{32}\ \sup_{1\le h\le m}\ \min\big\{\zeta h,\ \sigma h^{-1/2}\big\}.$$

Taking the maximum of the constant-path and ramp bounds proves the displayed lower bound. For the cube-root regime, assume $1 \le h_\star := (\sigma/\zeta)^{2/3} \le m$. Choose an integer $\bar h \in [h_\star/2, h_\star]$. Then

$$\zeta\bar h \ge \frac{1}{2}\,\zeta h_\star = \frac{1}{2}\,\sigma^{2/3}\zeta^{1/3},$$

and

$$\sigma\bar h^{-1/2} \ge \sigma h_\star^{-1/2} = \sigma^{2/3}\zeta^{1/3}.$$

Therefore

$$\min\big\{\sigma\bar h^{-1/2},\ \zeta\bar h\big\} \ge \frac{1}{2}\,\sigma^{2/3}\zeta^{1/3},$$

and the ramp bound gives

$$R^*_m \ge \frac{3}{64}\,\sigma^{2/3}\zeta^{1/3}.$$

So one admissible explicit constant in the critical-window regime is $c = 3/64$.

Remark 2.7 (Measurement layer). Debiased Sinkhorn divergence serves as a practical measurement layer when the underlying geometry must be estimated from samples. That computational choice supports the tracking law, but the law itself is about the horizon of useful memory, not about any particular OT solver.

3 Empirical Validation

The experiments test a single claim: under continuous drift, memory must be regulated by temporal validity rather than by statistical evidence of abrupt change alone. Empirically, the paper separates three layers that are often collapsed: horizon geometry, horizon tracking, and predictive payoff. The Gaussian sweep and lag–variance frontier validate geometry. The continuous-drift, temporal-validity, and alternating-timescales benchmarks probe tracking. The horizon cost curve assesses predictive payoff. A compact piecewise benchmark is retained only as a phase-wise calibration check.

3.1 Law Validation

We begin with the Gaussian sweep used to validate the finite-memory law. The setup produces the characteristic U-curve: for small windows, error is dominated by variance; for large windows, error is dominated by staleness. The empirical minimum follows the predicted cube-root scaling, and EWMA behaves as the natural finite-memory analogue rather than as a competing theory.

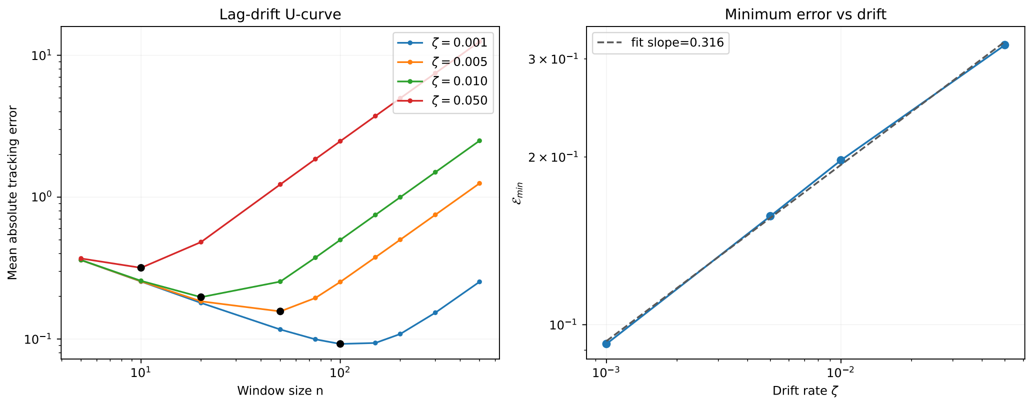

Figure 1: Cube-root validation on the Gaussian sweep. Left: tracking error as a function of window size, showing the predicted U-shape for multiple drift rates. Right: the minimum achievable error follows an approximate $\zeta^{1/3}$ law. This figure validates the structural trade-off between variance and staleness that underlies the rest of the paper.

3.2 Lag–Variance Frontier

The U-curve becomes more informative when viewed as a frontier in horizon space. The next figure plots tail error against effective memory horizon for a fixed-window sweep and overlays the operating points of the main methods. The curve is asymmetric: on the left, extra memory still buys visible noise reduction; on the right, extra memory mainly adds age. The knee is where that trade stops paying off.

3.3 Over-Memory Under Continuous Drift

The next benchmark focuses on the operational failure mode. We compare five methods on a smooth drifting stream: a short fixed window, a long fixed window,

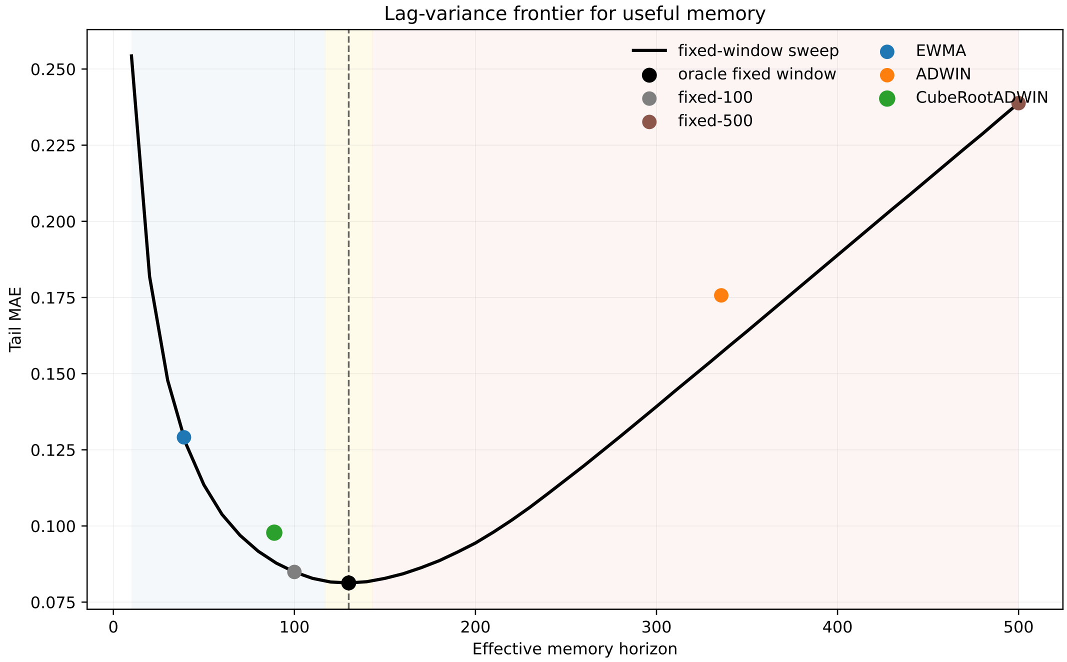

Figure 2: Lag–variance frontier for useful memory. The fixed-window sweep traces the empirical frontier: short horizons are noisy, long horizons are stale, and the knee marks the useful-memory optimum. The frontier is asymmetric because the gain from more memory saturates while the cost of age keeps accumulating. ADWIN sits far to the stale side, whereas CubeRootADWIN lies close to the frontier knee.

EWMA, ADWIN, and the cube-root capped variant. The question is when memory becomes stale before changepoint evidence accumulates.

Table 1: Continuous-drift benchmark under a smooth drift rate. The key gap is between statistical evidence and temporal validity: ADWIN keeps a much wider window than the cube-root regulator, while the long fixed window collapses under lag.

Method	Tail MAE	Mean horizon	Drift events	Cap events
Fixed-100	0.0849	100.0	–	–
Fixed-500	0.2388	500.0	–	–
EWMA	0.1291	39.0	–	–
ADWIN	0.1757	335.8	3.9	0.0
CubeRootADWIN	0.0978	88.9	0.0	1374.7

The key phenomenon is the separation between cap activation and the Hoeffding test. The cap can activate while the test remains silent, which makes memory validity distinct from the evidence threshold for abrupt change.

3.4 Temporal Validity Gap

The gap becomes more explicit when the drift rate varies. The next figure plots the first cap activation of the cube-root regulator and the first changepoint decision

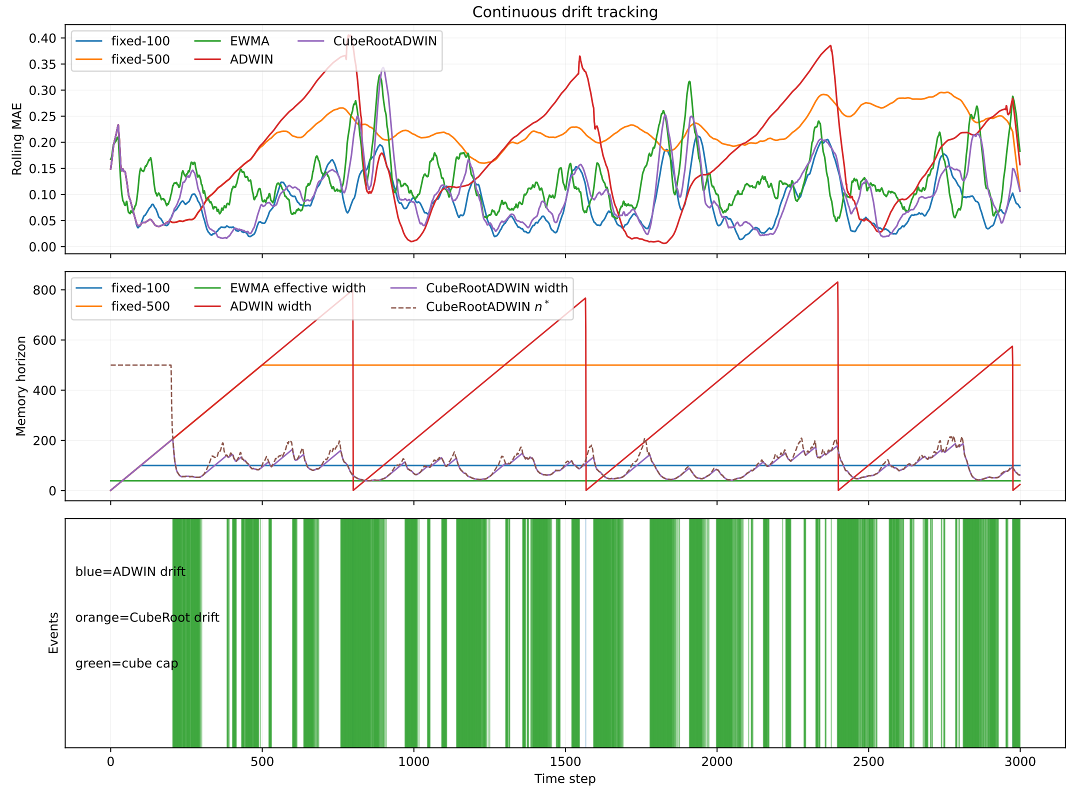

Figure 3: Continuous-drift benchmark. Fixed-500 is the stale-memory baseline; ADWIN retains far more memory than the oracle-scale horizon; CubeRootADWIN pulls the effective horizon toward the useful-memory scale and exposes a 'cap-only' regime, where memory becomes operationally stale before statistical changepoint evidence appears.

of ADWIN as the drift rate changes. In the smooth-drift regime, the cap activates much earlier than the detector; as the drift becomes faster, the gap shrinks but does not disappear.

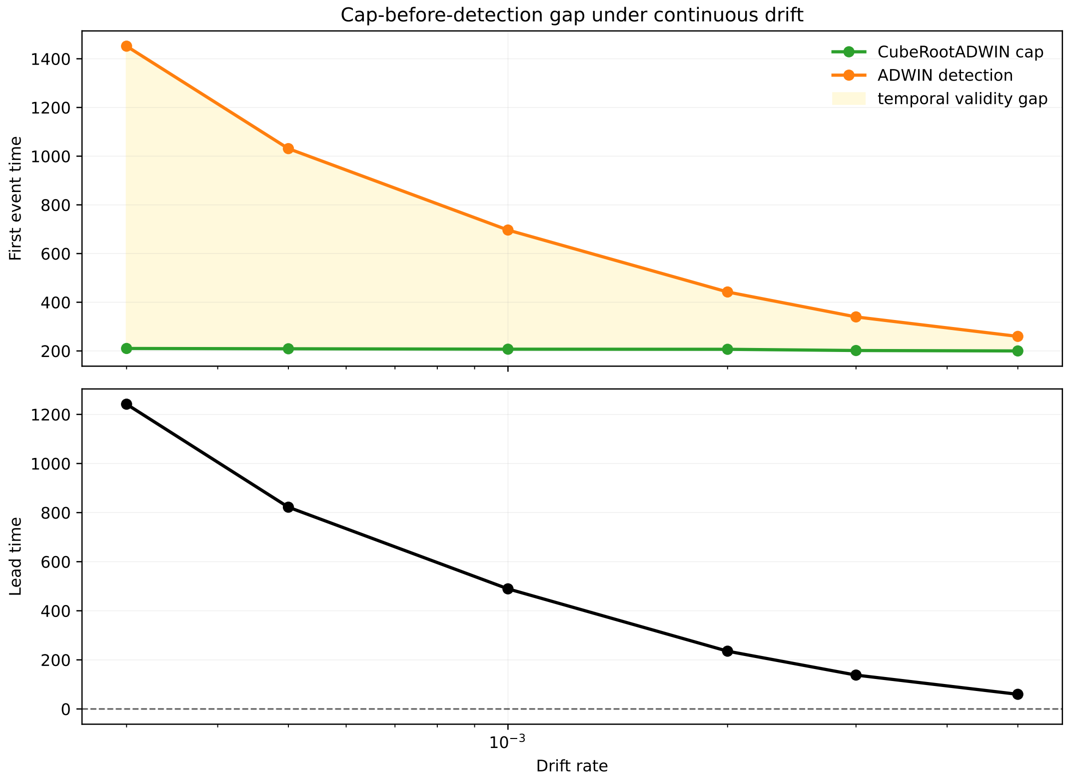

Figure 4: Temporal validity gap across drift rates. The top panel compares the first cube-root cap activation with the first ADWIN detection. The bottom panel shows the resulting lead time. Under slow drift, memory becomes operationally stale long before changepoint evidence is available; under faster drift, the gap narrows but remains positive.

3.5 Fixed-Horizon Instability

The next benchmark makes fixed-horizon instability visible. The stream alternates between slow drift, fast bursts, recovery plateaus, and micro-bursts with irregular durations, so the local oracle horizon need not be globally stable. In that setting, a single fixed window can look reasonable on average while remaining locally misaligned for long stretches. This benchmark separates three questions that are often conflated: whether the oracle horizon is time-varying, whether an online regulator can track that changing scale, and how far each method remains from the local oracle horizon over time.

This benchmark makes two things visible. First, horizon validity can vary even when aggregate prediction error stays competitive. Second, the adjustment is asymmetric: contraction of the useful horizon is easier to trigger than re-expansion after a low-drift stretch. The pattern is operational evidence of path dependence in temporal memory adaptation.

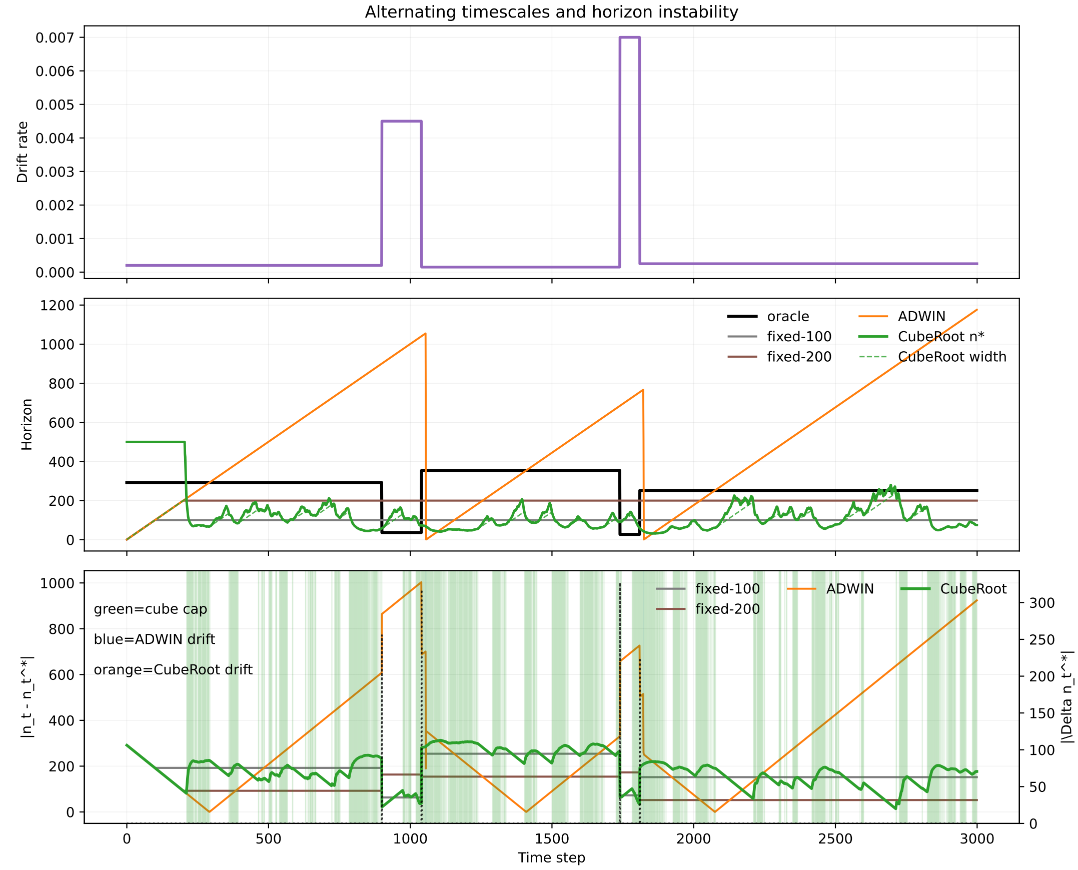

Figure 5: Alternating-timescales instability. The top panel shows the drift schedule, the middle panel shows the local oracle horizon alongside fixed windows and the cube-root regulator, and the bottom panel shows horizon regret $|n_t - n_t^*|$ over time. The purpose of the benchmark is to expose when the useful horizon changes locally: slow stretches favor longer memory, bursts contract the oracle scale, and fixed horizons can remain misaligned across regime changes. The figure makes the temporal-validity problem visible directly.

Table 2: Mean horizon regret over the first 50 steps after each regime change. The transition-window ablation shows that contraction and re-expansion are not symmetrical. Lower is better.

Method	Contraction	Expansion
fixed-100	63.3	254.2
fixed-200	163.3	154.2
ADWIN	888.8	435.8
CubeRoot	58.4	307.7

3.6 Horizon Cost Curve

The next question is when horizon misalignment becomes operationally expensive. For each backend, we compare its instantaneous horizon n_t with the local cube-root target n_t^* and plot the excess error $E_t - E_t^\star$ against both the absolute gap $|n_t - n_t^*|$ and the relative gap $|\log(n_t/n_t^*)|$. Here $E_t^\star$ is the best error achieved at time t by the fixed-window oracle grid, and the curve is binned over all seeds and time points.

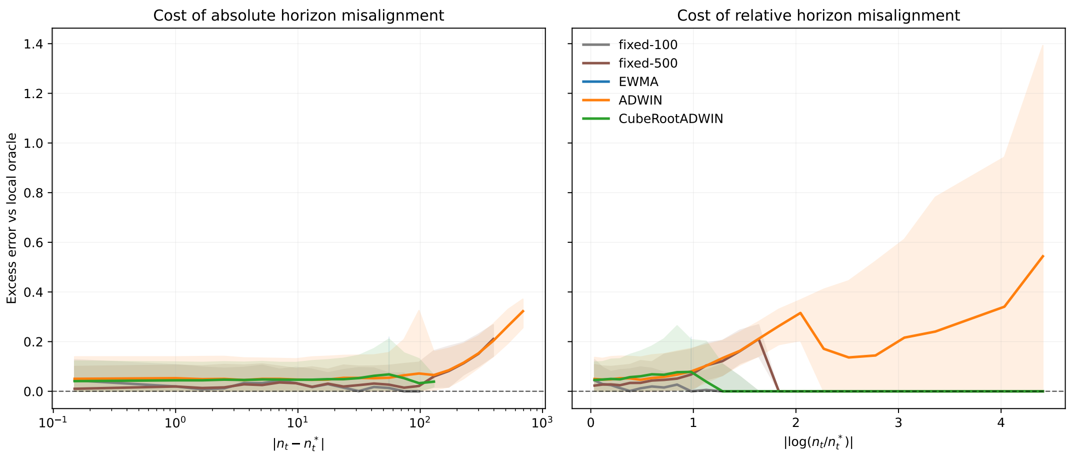

Figure 6: Excess error as a function of horizon misalignment. The relation is backend dependent rather than universal: fixed-100 stays inside a low-cost band, whereas fixed-500 and ADWIN pay substantially more once the gap widens. The cube- root regulator sits between these extremes.

The main lesson is that the paper exposes a measurable tolerance band around the useful-memory scale, together with backend-specific degradation once that band is exceeded.

3.7 Phase-Wise Recovery Check

The piecewise-drift benchmark changes the local drift rate in three phases, and the adaptive horizon is compared with the offline oracle fixed window for each phase. The calibration is held fixed across phases; only the stream changes. The detailed calibration check is reported in appendix A. In the main text, the point is simply that one calibration remains in the right operational range when the drift regime changes discretely.

3.8 What the Figures Show

Taken together, the figures support three claims. First, the cube-root law is a real geometric law for useful memory rather than a fitting artifact. Second, horizon tracking is distinct from changepoint detection: memory can become stale before statistical evidence is strong enough to declare a change. Third, horizon control has operational value, but its predictive payoff depends on the backend and on the drift regime.

4 Related Work

Tracking under nonstationarity. Dynamic regret, online least-squares, and adaptive filtering all ask a closely related question: how much past evidence should remain relevant once the target is moving [Besbes et al., 2015, Cheung et al., 2019, Yuan and Lamperski, 2020, Mazzetto and Upfal, 2023]. The present paper keeps that question in distributional form and makes the memory horizon itself the main object of analysis.

Transport geometry. Wasserstein distance and Sinkhorn-type estimators are natural because they respect the geometry of the signal space and behave well under transport of mass and support. Recent finite-sample results clarify how the constants depend on geometry and regularization, while leaving the exponent structure intact [Genevay et al., 2019, Niles-Weed and Bach, 2019, Goldfeld et al., 2024, Akyildiz et al., 2024]. Here, the geometry quantifies how much old evidence can be retained before it becomes stale.

Adaptive forgetting and horizon control. Classical smoothers such as EWMA, forgetting-factor RLS, and Kalman filtering all instantiate the same tension between variance reduction and responsiveness. What they often leave implicit is the temporal validity of the retained memory. The cube-root law makes that validity visible by turning the coherence-delay trade-off into an operational horizon selector.

Keehan, Anderson, and Wiesemann [Keehan et al., 2025] study a closely related drift-aware weighting problem in Wasserstein distributionally robust optimization. Their result balances effective sample size against drift in the weighted empirical distribution. The present paper differs in object and goal: it asks when memory remains temporally valid, rather than how to optimize a robust weighting rule.

Freshness and timeliness. The Age-of-Information literature gives a useful interpretive bridge: recent information is more timely, and timeliness carries a cost. That viewpoint fits the argument here, while AoI remains an interpretation rather than a modeling target.

What is different here. The result sits next to drift detection, but its organizing distinction is a different one: statistical evidence of change versus the horizon at which the retained memory remains current.

5 Limitations and Open Directions

Scope of the law. The cube-root law is exact for the finite-memory averaging class analyzed in this paper, and the lower bound is stated for window-restricted estimators on a W_2-Lipschitz drift class. It captures a structural finite-memory phenomenon, though not a universal minimax theorem over all possible online estimators.

Dependence on local drift estimation. The operational regulator depends on a local estimate of drift. The current benchmark suggests that the remaining gap to the oracle comes mostly from drift estimation. Better local geometry estimators may therefore sharpen the horizon recovery without changing the law.

Geometry beyond the current metric. The present formulation uses Wasserstein geometry and a debiased Sinkhorn measurement layer when samples must be compared computationally. Other geometries may admit analogous trade-offs, but the constants and the calibration of useful memory may change.

What the experiments show. The current experiments are not a leaderboard over drift detectors. They show a specific operational distinction: a tracker can still have statistically insufficient changepoint evidence while already being stale in the sense of useful memory.

6 Conclusion

How long can memory remain useful when the target distribution keeps moving? Under the drift class studied here, useful memory has a finite temporal horizon, and that horizon obeys a cube-root law. The governing object is a coherence- delay tradeoff: preserving temporally coherent evidence requires limiting the delay induced by old memory.

The theoretical contribution is a decomposition of tracking error into a variance term and a staleness term, together with a lower bound showing that the resulting finite-memory floor is structural. The operational contribution is a memory regulator: a cube-root cap keeps the effective horizon near the oracle scale across drift regimes, while classical adaptive windows can still accumulate stale memory.

The strongest empirical lesson is that horizon geometry, horizon tracking, and predictive payoff are distinct objects. Horizon misalignment carries a nonlinear operational cost, but the transfer from gap to error depends on the backend and on the drift regime.

Empirically, the signature regime is 'cap-only': the memory horizon becomes operationally stale before there is enough statistical evidence for a changepoint alarm. That separation between evidence of change and temporal validity of memory is the main conceptual payoff of the paper. The alternating-timescales benchmark adds a second operational lesson: horizon contraction is typically easier to induce than horizon re-expansion, revealing an asymmetric form of memory hysteresis.

Finite-memory systems are best judged by how long their retained evidence remains current enough to matter. The paper contributes a law with a clear design consequence: regulate useful memory and track the age at which evidence stops being current.

That is the paper's final claim: useful memory under drift has geometry, but its predictive value is policy-dependent.

Code and reproducibility artifacts are available in the project repository [Parreira, 2026].

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A Phase-Wise Recovery Details

This appendix presents the phase-wise calibration check for the cube-root regulator. The result is that the same calibration stays in the right operational range when the drift regime changes discretely.

Table 3: Oracle horizon recovery on piecewise drift. The same calibrated cap tracks the oracle scale across regimes: it is close in the fast-drift phase and remains within the right order of magnitude in the smoother phases.

Drift	Oracle window	Cube n^*	Oracle MAE	Cube MAE
0.0005	200	191.2	0.0741	0.0874
0.003	50	58.1	0.1258	0.1381
0.001	100	89.7	0.0876	0.1033

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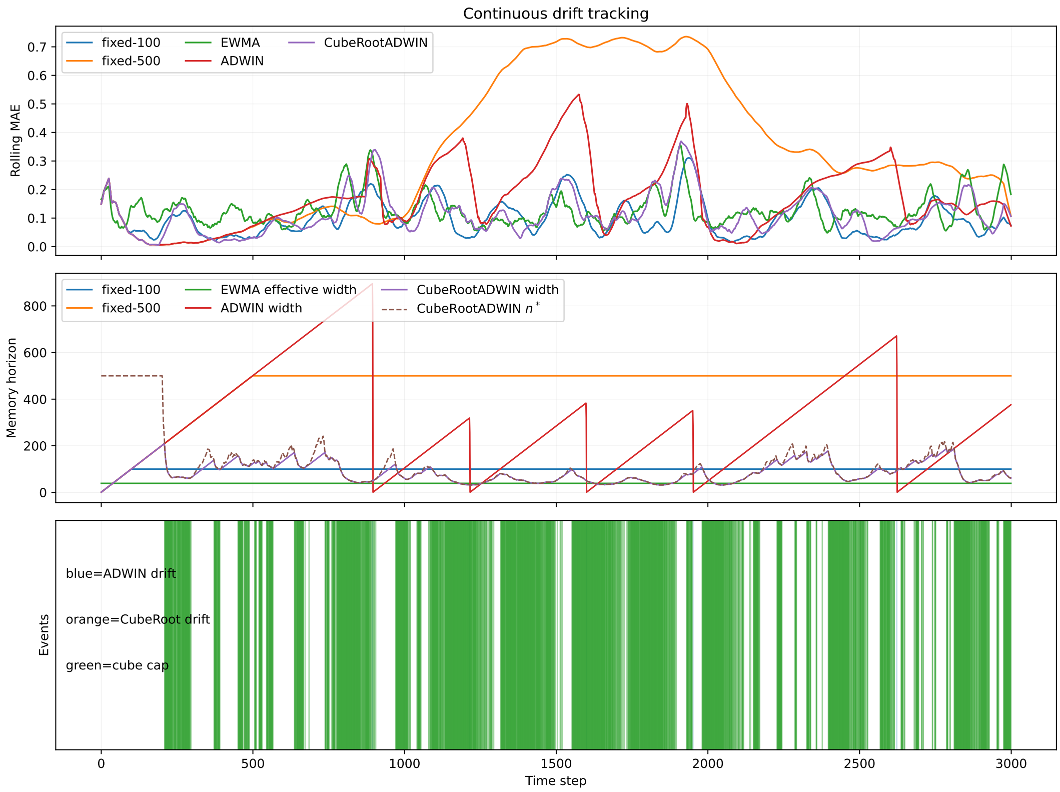

Figure 7: Piecewise-drift recovery. The same cube-root calibration follows the phase-dependent oracle window closely enough to preserve the right operational scale in each regime.

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